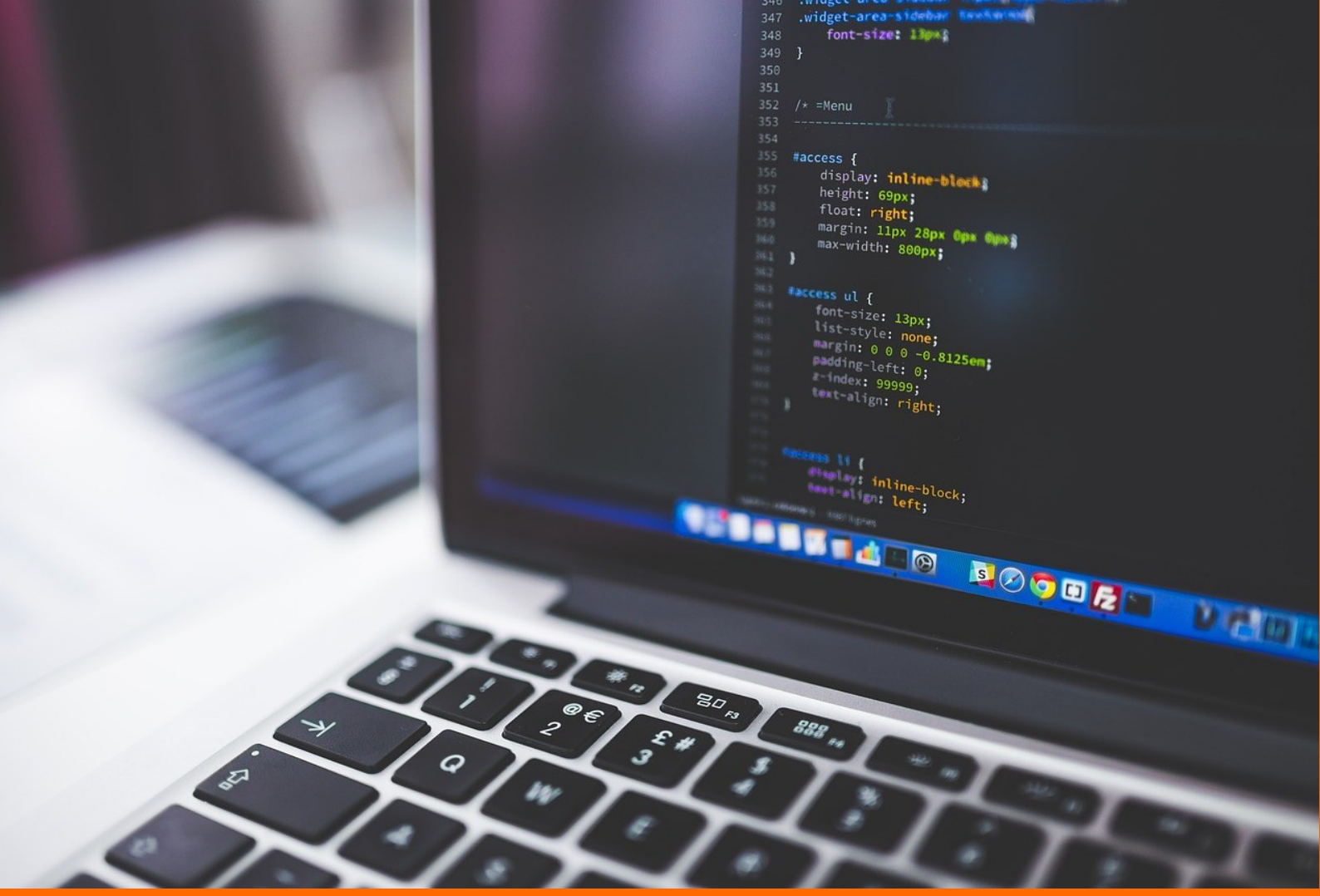

Development Specification Document

UiPath

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I. PURPOSE

The purpose of this Development Support Document (DSD) is to record the outcome specific to the automated master project and its sub-components: projects, workflows, and sequences.

Some of the content is taken from the Solution Design Document (SDD), while the content specific to this document is more detailed and is designed to enable troubleshooting of the automated solution during UAT as well as post-release into production.

This document serves as the primary operational reference for:

The RPA Developer (Zinedine Benmiya) during development and maintenance

Support developers who need to troubleshoot, fix, or enhance the automation

The business owner who needs to understand how to restart or recover from errors

Future team members taking over maintenance of the automation

1 AUTOMATED PROCESS DETAILS

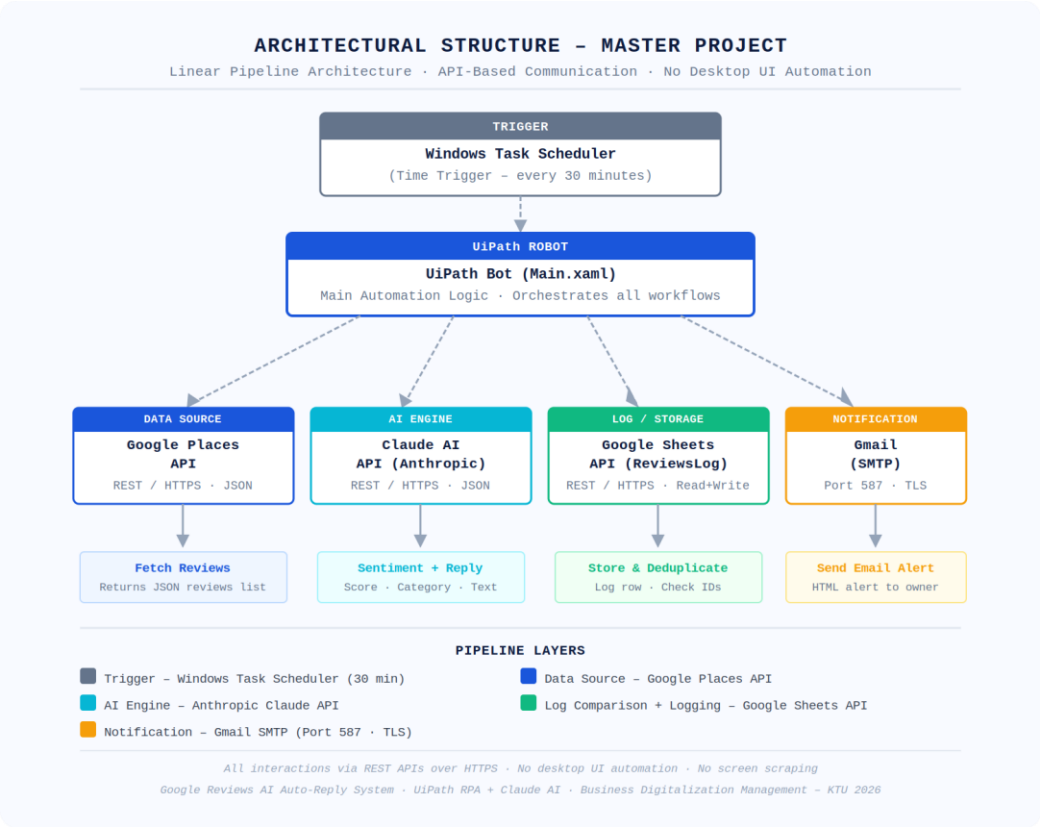
Details filled in need to reflect the actual information for the Master Project released for production. The following table will be populated:

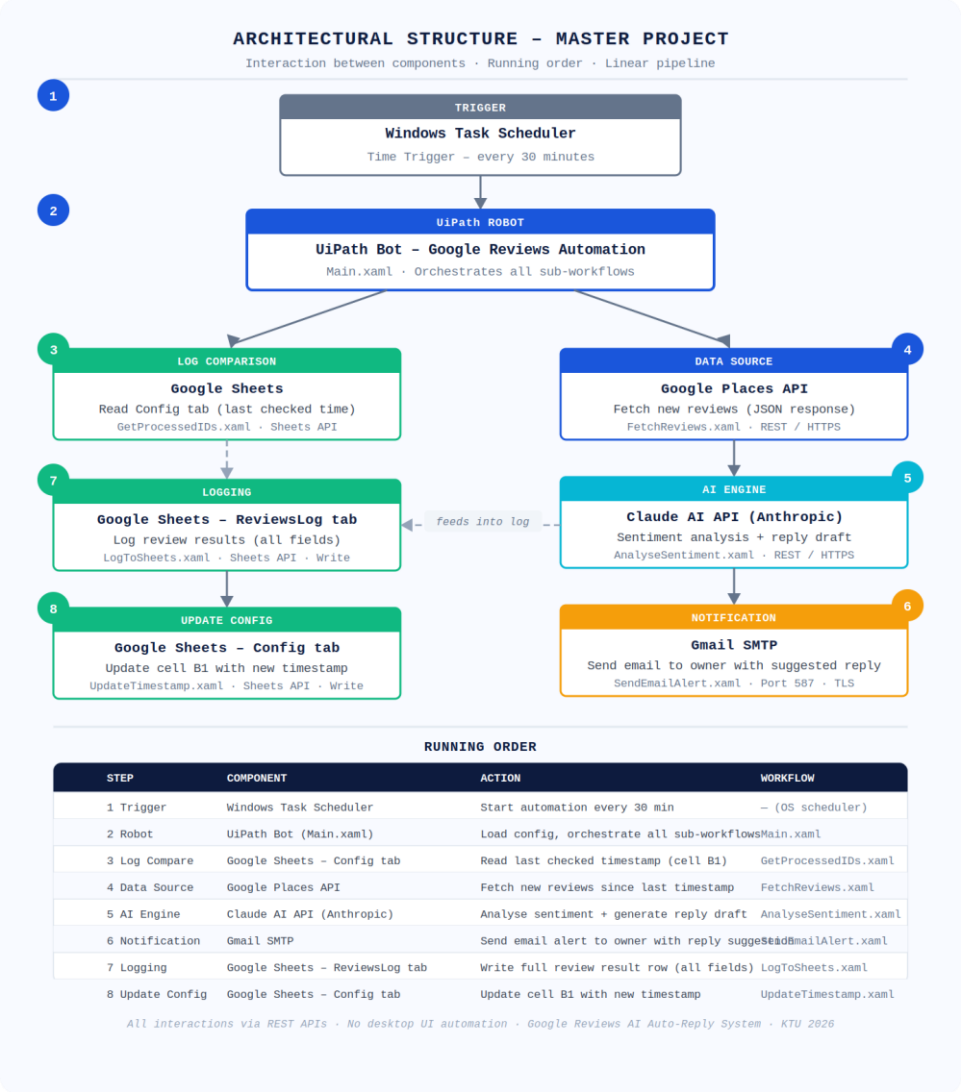
Item	Description
Master Project Name	Google Reviews Monitoring & AI Response Automation
Robot Type	Unattended (BOR – Back Office Robot)
Orchestrator used?	No – scheduled via Windows Task Scheduler (every 30 min)
Scalable	Yes – can monitor multiple Google Business locations
UiPath version used	UiPath Studio 2026.0.189 (Community Edition)

2 RUNTIME GUIDE

2.1 Architectural structure of the Master Project

The diagrams below display the interaction between components, packages, robots, and running order of the Master Project:





2.2 Master Project Runtime Details

Project is explained in higher detail in the table below:

ITEM NAME	DESCRIPTION **Fill in each bolded section - empty fields are not allowed. If the section does not apply to your automation then mark as n/a.**
Production environment details	*Running on Zinedine Benmiya's Windows 11 laptop (Automation Consultant, Kaunas, Lithuania). Scheduled every 30 minutes via Windows Task Scheduler. Machine must remain powered on and connected to the internet.*
Prerequisites to run	*• Internet connection active at all times • appsettings.json populated with valid API keys • Google Sheets spreadsheet accessible (service account has edit rights) • Gmail App Password*

	configured in appsettings.json • UiPath Robot runtime installed • Windows Task Scheduler task enabled and active
Input Data	*Google Business reviews (text, rating, author, timestamp) retrieved via Google Places API. Timestamp checkpoint from Google Sheets Config tab (cell B1) used to filter only new reviews.*
Expected Output	*• 1 HTML email alert sent to z.benmiya@gmail.com per new review containing full review details and AI-generated reply suggestion • 1 new row appended to Google Sheets ReviewsLog tab per processed review • Google Sheets Config tab B1 updated with new timestamp after each cycle*
How to start the automated process	*Process starts automatically via Windows Task Scheduler every 30 minutes. For manual start: Open UiPath Studio → Open Main.xaml → Click Run. Alternatively via command line: uipath run GoogleReviews_AI_AutoReply*
How to restart the process from a certain action?	*The process cannot be restarted from a specific step. It must be restarted from the beginning (Main.xaml). However, since the robot uses the Google Sheets Config tab B1 timestamp as a checkpoint, resetting B1 to an earlier timestamp will cause the robot to reprocess reviews from that point onwards on the next run.*
Reporting (queues reporting, Kibana or another platform)	*• Google Sheets – ReviewsLog tab: full audit log of all processed reviews • Google Sheets – ErrorLog tab: all API errors, retry counts, failure reasons • Gmail inbox: HTML email alert per new review • Local backup: /Data/Backup/backup_reviews.csv (Sheets API fallback) Live Audit Log: https://docs.google.com/spreadsheets/d/1rrSW-JNwvnfElAsSGJ-Ks6Pwcdop-sHRE_vn-0ZdlNA/edit?usp=sharing*
Manual error handling? (roll back or manually complete failed transactions)	*In case of API error (Google Places, Claude AI, or Sheets), the robot automatically retries 3 times with a 10-second delay before logging to ErrorLog tab. No manual rollback is required. Failed reviews are simply not logged — they will be reprocessed on the next cycle if their timestamp is newer than Config tab B1. To manually trigger reprocessing of a specific review: update Config tab B1 to a timestamp before that review was posted.*
How to resume the process in case of error?	*1. Check the ErrorLog tab in Google Sheets to identify the error type and workflow that failed 2. Verify all API keys in appsettings.json are valid and have sufficient credits 3. Check internet*

	connectivity 4. Verify Google Sheets is accessible and service account has edit permissions 5. Re-run Main.xaml manually or wait for the next Task Scheduler trigger (max 30 min) 6. Monitor the ReviewsLog tab to confirm reviews are being processed again
How to manually fix transactions with error?	*If a review was not processed due to an API failure: 1. Open Google Sheets ErrorLog tab and note the review ID and error details 2. Fix the root cause (e.g. renew API key, restore internet connection) 3. Update Config tab B1 to a timestamp before the failed review to trigger reprocessing 4. The robot will process it on the next scheduled run If email alert was not sent: check Gmail SMTP credentials in appsettings.json and verify Gmail App Password is still valid.*
How is Orchestrator used?	*Not used in Phase 1. Scheduling is handled entirely by Windows Task Scheduler. UiPath Orchestrator is planned for Phase 2 to provide centralised scheduling, monitoring, and multi-robot management.*
Password policies (mention any specific compliance requests)	*• Gmail App Password (16-character) – not expiring; generated from Google Account Security settings • All API keys (Google Places, Google Sheets service account, Anthropic Claude) stored in appsettings.json – gitignored • No credentials hardcoded in any workflow file • Claude API key should be rotated annually*
Stored credentials (Never use hardcoded credentials in the workflow!)	*Stored in /Data/Config/appsettings.json on the local machine. File is listed in .gitignore. Template provided as appsettings.example.json in the GitHub repository. Service account JSON file stored separately at path defined in appsettings.json.*
List of Asset Names (Naming convention: ProcessName_AssetName)	*n/a – No UiPath Orchestrator assets used in Phase 1. All configuration stored in local appsettings.json.*
List of queues names (Naming convention: ProcessName_QueueName)	*n/a – No UiPath Orchestrator queues used in Phase 1. Reviews processed directly from Google Places API JSON response filtered by timestamp.*
Schedule Details	*Every 30 minutes, 24/7 – triggered by Windows Task Scheduler. No peak restrictions. Runs between any hours and on weekends.*
Multiple Resolutions Supported? (in case of image automation / Citrix and VDI)	*n/a – No image automation or UI interaction. All operations are API-based (REST over HTTPS). Screen resolution is irrelevant.*
Recommended Resolution	*n/a – No UI automation used.*

2.3 Project details

In this section describe all the projects that compose the automated process. For each project the following table should be filled in.

Project Name: Taken from Solution Architecture Document

ITEM NAME	DESCRIPTION Fill in each section - empty fields are not allowed. If the section does not apply to your automation then mark as n/a.
Environment used for development (name, location, configuration details etc)	DEV_Local_WIN11 – Zineb Benmiya's Windows 11 laptop, Kaunas, Lithuania. UiPath Studio 2026.0.189 (Community Edition). Development and production run on the same machine
Environment prerequisites (OS details, libraries, required apps)	Windows 11 (64-bit) • UiPath Studio 2026.0.189 Community Edition • .NET 6.0 Runtime or higher • UiPath Robot runtime • Internet access (required for all API calls) • Google Cloud project with Places API and Sheets API enabled
Repository for project (where is the developed project stored)	GitHub: github.com/zinedine-benmiya/GoogleReviews_AI_AutoReply Local path: C:\Users\Zinedine\Projects\GoogleReviews_AI_AutoReply\
Configuration method (assets, excel file, Json file)	JSON file – /Data/Config/appsettings.json Contains: GooglePlacesApiKey, GooglePlaceId, AnthropicApiKey, GoogleSheetsId, GoogleServiceAccountJson path, GmailFrom, GmailAppPassword, GmailTo, PollingIntervalMinutes, MaxRetries, RetryDelaySeconds
Configuration details (path for input files, configuration file, Orchestrator assets used)	• Config file: /Data/Config/appsettings.json • Service account: /Data/Config/service-account.json (gitignored) • Email template: /Data/Templates/email_template.html • Backup log: /Data/Backup/backup_reviews.csv • Exception screenshots: /ExceptionScreenshots/ • No Orchestrator assets used
List of reused components	n/a – No components reused from existing automations or Automation Hub. All workflows developed from scratch.
Logging Level	Info – all key process events logged (review detected, sentiment result, email sent, row written to Sheets). Error level used for API failures and retry attempts.
Details about automation (if	No UI automation used at all. The entire automation is API-based: • Google Places API (REST/HTTPS GET) – fetches reviews • Anthropic

the apps were automated using UI Automation, Image & Text)	Claude API (REST/HTTPS POST) – sentiment analysis • Google Sheets API (REST/HTTPS GET/POST) – read Config tab, write ReviewsLog • Gmail SMTP (Port 587/TLS) – send email alerts No selectors, no image recognition, no screen interaction.
In case of FOR, can the user operate the computer while the robot is running?	n/a – This is a BOR (Back Office Robot / Unattended). The robot runs completely in the background. The user can freely operate the computer while the robot is running, as there is no UI interaction.
Custom error logs defined in the workflows (where Throw Activity was used or custom log message was defined)	1. PlacesAPI_NoReviews – Google Places API returned empty reviews array for the business location 2. PlacesAPI_RateLimitHit – Google Places API returned HTTP 429 (rate limit exceeded) 3. ClaudeAPI_Timeout – Anthropic Claude API did not respond within 30 seconds 4. ClaudeAPI_InvalidKey – Anthropic API returned HTTP 401 (invalid or expired API key) 5. SheetsAPI_WriteFailed – Google Sheets API returned error when appending ReviewsLog row 6. SheetsAPI_ReadFailed – Google Sheets API failed to read Config tab B1 timestamp 7. SMTP_ConnectionFailed – Gmail SMTP connection refused on Port 587 8. Config_MissingKey – Required key missing from appsettings.json at startup
Frequent errors found in the development phase	1. Google Places API rate limiting (HTTP 429) when testing with many rapid calls – resolved by adding 60s wait before retry 2. Claude AI reply quality inconsistent on very short reviews (e.g. '5 stars') – resolved by improving prompt to handle rating-only reviews 3. Google Sheets service account permission denied – resolved by sharing the spreadsheet with the service account email address 4. Gmail App Password rejected – resolved by ensuring 2FA is enabled on the Gmail account before generating App Password 5. Config tab B1 timestamp format mismatch causing all reviews to be reprocessed – resolved by standardising to ISO 8601 format
Workarounds used in the automation phase	1. Google Places API instead of GBP API for fetching reviews – GBP API was applied for but rejected by Google; Places API provides review data sufficiently for Phase 1 2. Gmail SMTP instead of Gmail API – simpler to configure, no OAuth flow required, App Password sufficient for Phase 1 3. Timestamp-based deduplication instead of review ID tracking – Config tab B1 stores last run timestamp to filter new reviews without needing a full ID lookup on every cycle 4. Local CSV fallback when Google Sheets API fails – backup_reviews.csv ensures no data loss during Sheets outages

2.4 Project(s) workflows

Workflows specific to: GoogleReviews_AI_AutoReply

Workflow Name	Description	Input Argument	Output Argument
Main.xaml	Entry point. Reads config, invokes all sub-workflows in sequence, handles global exceptions and loop logic.	configPath (String)	n/a (orchestrator)
FetchReviews.xaml	Calls Google Places API with API key and Place ID. Returns list of all review objects.	in_PlacesApiKey (String) in_PlaceId (String)	out_Reviews (List<JObject>) out_Success (Boolean)
GetProcessedIDs.xaml	Reads Google Sheets Config tab B1 timestamp. Returns list of already-processed review IDs for deduplication.	in_SheetsId (String) in_ServiceAccountJson (String)	out_ProcessedIDs (List<String>) out_LastTimestamp (DateTime)
AnalyseSentiment.xaml	Calls Claude AI API with review text. Returns Sentiment (Positive/Neutral/Negative), Score (0-100), and SuggestedReply.	in_ReviewText (String) in_ReviewRating (Integer) in_ClaudeApiKey (String)	out_Sentiment (String) out_ConfidenceScore (Integer) out_SuggestedReply (String) out_Success (Boolean)
LogToSheets.xaml	Writes review data and AI analysis result to Google Sheets ReviewsLog tab via Sheets API.	in_ReviewId, in_ReviewerName, in_Rating, in_ReviewText, in_Sentiment, in_ConfidenceScore, in_SuggestedReply, in_SheetsId	out_Success (Boolean)
SendEmailAlert.xaml	Sends formatted HTML email to owner via Gmail SMTP with review details and AI reply suggestion.	in_ReviewerName, in_Rating, in_ReviewText, in_Sentiment, in_ConfidenceScore,in_SuggestedReply, in_GmailFrom, in_GmailTo, in_GmailPassword	out_Success (Boolean)
UpdateTimestamp.xaml	Writes current datetime to Google Sheets Config tab B1 after successful cycle completion.	in_SheetsId (String) in_Timestamp (DateTime)	out_Success (Boolean)

ErrorHandler.xaml	Centralised exception handler. Retries 3x with 10s delay, then logs to ErrorLog tab and sends error alert email to owner.	in_ExceptionMessage (String) in_WorkflowName (String) in_RetryCount (Integer)	out_ShouldRetry (Boolean) out_FinalError (Boolean)

3OTHER DETAILS

Any other points relevant to the automation process:

Debugging Tips

1. Always check the Google Sheets ErrorLog tab first , it contains the exact error message, the workflow that failed, and the retry count. This is the fastest way to diagnose any issue.

2. If the robot stops processing reviews, check Config tab B1. If the timestamp is in the future or corrupted, update it manually to the correct datetime in ISO 8601 format (e.g. 2026-04-12T10:00:00Z).

3. If Claude AI returns poor quality replies, check the in_ReviewText value being passed — very short reviews or non-English text may produce generic responses. Review the prompt in AnalyseSentiment.xaml.

4. If Google Sheets write fails repeatedly, verify the service account email has Editor access to the spreadsheet. Open the spreadsheet → Share → confirm the service account email is listed with Editor role.

5. If Gmail SMTP fails, verify the Gmail App Password in appsettings.json is exactly 16 characters with no spaces. Regenerate if needed from Google Account → Security → App Passwords.

6. If the Places API returns no reviews unexpectedly, verify the GooglePlaceId in appsettings.json is correct by testing it manually at: https://maps.googleapis.com/maps/api/place/details/json?place_id=YOUR_ID&key=YOUR_KEY

7. Exception screenshots are automatically saved to /ExceptionScreenshots/ when unhandled exceptions occur, check this folder when diagnosing unexpected crashes.

Post UAT Specifications

ITEM NAME	DESCRIPTION
Average Handling Time per Transaction	Approximately 8–15 seconds per review (includes Google Places API call, Claude AI analysis, Google Sheets write, and email send). Total cycle time depends on the number of new reviews; typically completes in under 2 minutes for up to 10 new reviews.
Recommended Number of Robots for the specified volumes (volumes captured during Assessment and PDD)	1 robot (BOR) is sufficient for the current volume of Automation Consultant (estimated 0–10 new reviews per day). If volume increases significantly (50+ reviews/day), consider parallelising review processing or upgrading to UiPath Orchestrator with multiple robots.
Specified Schedule	Every 30 minutes, 24/7 – triggered by Windows Task Scheduler. No peak restrictions. Consistent schedule regardless of day or time.

4GLOSSARY

The main terms used in the Solution Architecture Document are defined below:

Master project - The overall output of the development — GoogleReviews_AI_AutoReply — containing all workflows that cover the full scope of the robotic process automation. 1-to-1 connection with the process described in the PDD.

Project - A UiPath Studio project containing one or multiple workflow files. Can be converted to a package and run independently.

Package - The output of compiling one or multiple projects. Deployed on the robot machine and executed by the UiPath Robot service. Only one package can execute at a time.

Activity - A component of the package saved as a .xaml file. Encapsulates part of the project logic. Can be of type: sequence, flowchart, or state machine. Invoked from Main.xaml.

Sequence - A workflow where activities are executed one after another in sequential order. Used in all sub-workflows of this project.

Flowchart - A workflow where activities are connected by arrows. Logic can be followed visually. Used in Main.xaml for the review processing loop.

State machine - A more advanced workflow organisation, similar to a flowchart. Not used in this project.

BOR - Back Office Robot – an unattended robot that runs without human interaction, triggered by a scheduler. This project uses a BOR triggered by Windows Task Scheduler.

FOR – Front Office Robot – an attended robot that requires human interaction. Not applicable to this project.

Orchestrator – Enterprise architecture server platform supporting release management, centralised logging, reporting, auditing, monitoring, remote control, scheduling, queue/robot workload management, and asset management. Planned for Phase 2.

Google Places API– A Google service used to retrieve Google Business reviews including review text, rating, author, and timestamp. Used by FetchReviews.xaml.

Claude AI– An AI model provided by Anthropic used for sentiment analysis and generating personalised reply suggestions. Used by AnalyseSentiment.xaml.

Sentiment Analysis– The process of classifying reviews as Positive, Neutral, or Negative based on their content, along with a confidence score (0–100).

ReviewsLog – Google Sheets tab where all processed reviews are logged. Columns: Review ID, Reviewer Name, Rating, Review Text, Sentiment, Confidence Score, Suggested Reply, Date Processed.

Timestamp (Checkpoint) – A datetime value stored in Google Sheets Config tab cell B1. Used to track the last processed review time so only new reviews are processed in each cycle.

Email Notification – An automated HTML email sent to the business owner via Gmail SMTP containing full review details and the AI-generated reply suggestion.

appsettings.json – Local configuration file at /Data/Config/appsettings.json containing all API keys and operational parameters. Gitignored — never committed to version control.